Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS at USC with 2+ years building healthcare-style portals, SQL-backed workflows, API integrations, and automation tools. Strong in JavaScript, Python, SQL, Web APIs, bug triage, Git, and ramping into .NET/Azure/EDI platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and optimized SQL-backed portal workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through query tuning, indexed access patterns, and service fixes.
- Built validation pipelines processing 5,000+ monthly jobs with reconciliation checks, CI/CD releases, and runbooks, improving reporting reliability and operational handoffs.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed React and Flask portal features handling 10,000+ daily requests, converting ambiguous user needs into responsive interfaces, REST APIs, tests, and incremental releases.
- Built automated document workflows with OCR, Python services, and AWS-backed persistence, reducing turnaround by **60%** while documenting extraction edge cases and integration failures.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built Python evaluation pipelines across 100,000+ records, producing reproducible reports, debugging artifacts, and visual summaries that clarified model behavior and data quality.
- Analyzed failure patterns across generated outputs and datasets, translating complex findings into actionable recommendations while strengthening validation coverage and technical decision-making.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and accelerating defect isolation across services.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

Web & APIs: JavaScript, TypeScript, HTML/CSS, React, Node.js, REST APIs, Web API patterns, C#/ASP.NET ramp-up
Data: SQL, PostgreSQL, MySQL, SQL Server familiarity, MongoDB, relational queries, reporting, data validation
Healthcare Ops: Claims/benefits workflow familiarity, portals, EDI learning, CRM integrations, workflow automation, issue triage
Cloud & AI: Azure learning, AWS, Git/GitHub, Docker, Jenkins, CI/CD, GitHub Copilot/Codex workflows, scripting

Projects

HMIS Code | *MERN, TypeScript, ABDM APIs, Sentry*

- Built a healthcare management platform integrating patient, provider, and claim workflows with secure APIs, structured validation, monitoring, and quality-oriented operational reliability.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an AI-powered SQL workspace supporting 10+ databases, combining schema inspection, query generation, Dockerized services, and debugging flows for faster data validation.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Built a secure document Q&A platform with semantic search, summarization, and prompt workflows over 50K+ records, reducing review time by **60%**.